

Mini-MRV - Model Validation Report (MVR)

Indo-Gangetic Wheat (Ludhiana + Karnal) | RothC-lite (not DayCent) | Verra VM0042 / VMD0053

Results - Mean and Conservative tCO2e per ha per year

District	Scenario	Mean	SD	P5	P95	Width	Deduct	Conservative
Karnal	CT_burn	-2.641	0.447	-3.348	-1.874	1.474	YES	-3.348
Karnal	ZT_FYM	-0.194	0.577	-1.098	0.774	1.871	YES	-1.098
Karnal	ZT_retain	-1.399	0.427	-2.109	-0.697	1.411	YES	-2.109
Ludhiana	CT_burn	-2.771	0.419	-3.436	-2.043	1.393	YES	-3.436
Ludhiana	ZT_FYM	-0.377	0.540	-1.210	0.537	1.747	YES	-1.210
Ludhiana	ZT_retain	-1.521	0.382	-2.152	-0.859	1.293	YES	-2.152

Methodology

- SOC: SoilGrids 250m OCS 0-30cm via ISRIC WebDAV (real). ~120 pixels per district, Ludhiana + Karnal.
- Model: RothC-lite $SOC(t+1)=SOC(t)+C_{input}*h - k*SOC(t)*f_{temp}*f_{moist}*f_{till}$; monthly, 20yr; h_burn 0.12, h_retain 0.20, h_FYM 0.32; k 0.032/0.028; f_till 1.35 CT vs 0.92 ZT.
- NDVI is a mock proxy (real: Planetary Computer STAC). $C_{retain}=(NDVI-0.30)*5.5$, $C_{burn}=retain*0.48$, $C_{FYM}=retain+0.85$.
- Credit = incremental_tco2e_vs_baseline (project minus CT_burn). Uncertainty: Monte Carlo 1000, $k \sim N(base, 0.006)$; VMD0053 deduction if 90% CI width > 50% of |mean|, conservative = p5.

Limitations (honest)

- No field SOC calibration; absolute tCO2e illustrative, incremental deltas are the valid signal.
- RothC-lite, NOT DayCent. No N2O/CH4. Real DayCent needs soil horizons + daily weather + field calibration.
- NDVI and climate modifiers are mock. Real: Planetary Computer STAC + ERA5 CDS.

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